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Movement Pattern Training To Improve Function in People with Chronic Hip Joint Pain: A Feasibility Randomized Clinic Trial

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Abstract

STUDY DESIGN—Feasibility randomized clinical trial

BACKGROUND—Rehabilitation may be an appropriate treatment strategy for patients with chronic hip joint pain (CHJP), however the evidence related to the effectiveness of rehabilitation is limited.

OBJECTIVES—Assess feasibility of performing a randomized clinical trial to investigate the effectiveness of movement pattern training (MPT) to improve function in people with CHJP.

METHODS—Thirty-five patients with chronic CHJP were randomized into two groups, treatment (MPT) or wait-list control (Wait-list). The MPT program included six, one hour supervised sessions and incorporated: 1) task-specific training for basic functional tasks and symptom-

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Statement of Financial Disclosure and Conflict of Interest

I affirm that I have no financial affiliation (including research funding) or involvement with any commercial organization that has a direct financial interest in any matter included in this manuscript, except as disclosed in an attachment and cited in the manuscript. Any other conflict of interest (ie, personal associations or involvement as a director, officer, or expert witness) is also disclosed in an attachment.

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provoking tasks; and 2) strengthening of hip musculature. The Wait-list group received no treatment. Primary outcomes for feasibility were patient retention and adherence. Secondary outcomes to assess treatment effects were patient-reported function (Hip disability and Osteoarthritis Outcome Score [HOOS]), lower extremity kinematics, and hip muscle strength.

RESULTS—Retention rates did not differ between MPT (89%) and Wait-list groups (94%, P = 1.0). Sixteen of the 18 patients (89%) in the MPT group attended at least 80% of the treatment sessions. For the home exercise program, 89% of patients reported performing their home program at least once per day. Secondary outcomes support the rationale for conduct of a superiority RCT.

CONCLUSION—Based on retention and adherence rates, a larger RCT appears feasible and warranted to assess treatment effects more precisely. Data from this feasibility study will inform our future clinical trial.

Keywords

femoroacetabular impingement; hip dysplasia; kinematics; movement system; strength

INTRODUCTION

Chronic hip joint pain (CHJP) due to conditions such as femoroacetabular impingement, hip dysplasia and labral tears, result in substantial hip pain and dysfunction in young adults. Patients most commonly report sharp or achy pain located deep in the hip joint or anterior groin.[6, 8, 12, 38] CHJP leads to limitations in walking, sitting and standing,[6, 8, 38] restricting the ability to work or perform everyday tasks. Without proper management, CHJP may progress to hip osteoarthritis (OA),[4, 17, 31, 44] a leading cause of reduced quality of life and loss of function for older people. Early, effective treatment of CHJP is needed to improve function in the young adult and prevent or delay the onset of hip OA.

Rehabilitation to address modifiable factors, such as abnormal movement patterns and hip muscle weakness, associated with CHJP could lead to improved function and reduced pain in people who have CHJP. An abnormal movement pattern such as medial collapse, where the lower limb collapses medially during weight bearing tasks, has been implicated as a risk factor for developing other musculoskeletal conditions such as patellofemoral pain[37, 41] and non-contact anterior cruciate ligament tears.[19] Specific to CHJP, excessive hip adduction associated with medial collapse movement pattern has been observed in people with CHJP[2, 25] and may be associated with articular cartilage damage in the hip joint.[25] Abnormal movement patterns may create altered mechanical forces on joint structures,[40] thus changing the location and magnitude of stress to specific joint tissues, such as the cartilage and acetabular labrum. Repeated loading of the joint with altered mechanical forces may contribute to cumulative tissue stress, micro-trauma, pain and potentially osteoarthritis.[13]

A second modifiable factor associated with CHJP is hip muscle weakness.[7, 16] Muscles that closely surround the hip joint are proposed to provide dynamic stability of the femur within the acetabulum.[39] In particular, the short external rotators (ERs),[46] abductors (ABDs)[34] and the iliopsoas[27] are all thought to contribute to dynamic stability of the joint. The ABDs and

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ERs may also assist in controlling excessive medial collapse during weight bearing activities such as gait and single leg squat. Previous studies have shown that people with CHJP have significant weakness in the hip ERs,[7, 16] ABDs[7, 16] and flexor muscles.[7] Rehabilitation to address abnormal movement patterns and muscle weakness may help to restore joint precision and dynamic stability, reduce excessive stresses and allow healing of injured tissues. We therefore developed a rehabilitation approach, movement pattern training (MPT), with the goal to reduce stresses on the hip joint by optimizing the biomechanics during functional tasks.

The purpose of this study was to assess feasibility of performing a randomized clinical trial to investigate the effectiveness of MPT to reduce symptoms and improve function in people with CHJP. We randomized patients with CHJP into two groups, a treatment group and a wait-list control group. The primary objectives of the study were to estimate 1) patient retention rates, 2) patient adherence to supervised treatment sessions and 3) patient adherence to home exercise program. The secondary objective was to compare the effects of MPT to no treatment on patient-reported function, lower extremity movement patterns and hip muscle strength.

METHODS

Study Design

This study was a feasibility randomized clinical trial (RCT). Figure 1 provides an overview of study design.

Participants

The patients in this study were a subset of participants from a prospective cohort study to assess risk factors for CHJP. For the parent study, we recruited patients with CHJP and asymptomatic participants to participate in a clinical examination, kinematic assessment and magnetic resonance imaging (MRI). Patients with CHJP also participated in the treatment component of the study. Recruitment for both the parent and the current study began in 2011 and ended in 2013. Potential participants, aged 18–40 years, were recruited from the Washington University's Orthopaedic, Physical Medicine and Rehabilitation, and Physical Therapy clinics, Washington University's research volunteer database and through public announcements. The age restriction ensured that participants were skeletally mature, but unlikely to have advanced age-associated degenerative joint disease. To be eligible for the parent study, patients had to experience CHJP as reported by deep hip joint or anterior groin pain lasting longer than 3 months that was reproducible with the Flexion-Adduction-Internal Rotation impingement test, also known as the FADIR or FAIR test.[28] The FADIR test can accurately detect hip joint-related pain[29] and pathology[33] with high test reliability.[29] Exclusion criteria included: 1) previous hip surgery or fracture, 2) contraindication to MRI, 3) known pregnancy, 4) neurological involvement that influenced coordination or balance, 5) knee or low back pain that limited mobility and 6) body mass index (BMI) greater than 30. Two exclusion criteria, contraindication to MRI and known pregnancy were required for testing procedures in the parent study. Neurological involvement and BMI exclusions were used to optimize kinematic testing. Additionally, patients were excluded if screening tests

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for differential diagnosis were positive for possible lumbar spine radiculopathy. This study was approved by the Human Research Protection Office of Washington University School of Medicine and all patients signed an informed consent statement prior to participating in the study.

Baseline Assessment

After informed consent was obtained, patients completed questionnaires to obtain demographic information and patient-reported outcomes prior to testing. A clinical examination by a trained physical therapist was then performed to obtain hip muscle strength followed by kinematic assessment to quantify lower extremity movement patterns. If a patient reported bilateral pain, the most painful hip was tested.

Randomization

Immediately after baseline assessment, patients were randomized to the treatment group (MPT) or the wait-list control group (Wait-list). A blocked randomization with a varying block size was used to ensure that there was appropriate concealment. Randomization sequences were generated a priori by the Research Design and Biostatistics Group (RDBG). The examiner performing the baseline and follow-up assessments was blinded to group assignment. Given the nature of the MPT intervention, it was not possible to blind the treating physical therapist or the patients to the treatment assignment; however, the treating physical therapist did not participate in outcome assessment.

After the baseline assessment, patients met independently with a research assistant who revealed the group assignment and assisted with scheduling. Patients randomized to the MPT group were scheduled to begin treatment within one week of their baseline assessment. The research assistant helped the patients schedule their first treatment session. Consistent with clinical procedures, the treating physical therapist was responsible for scheduling the remaining weekly appointments for a total of six sessions. Patients were then recontacted to schedule their follow-up testing session. Patients randomized to the Wait-list group were scheduled for their follow up testing session six weeks after baseline assessment. After participation in the follow-up testing session, patients in the Wait-list group were offered the same treatment using MPT.

Intervention

Movement Pattern Training—Movement pattern training consisted of six, one hour visits over six weeks. The duration of six weeks was based on our previous work,[21] and other reported clinical studies.[32, 47] Treatment for all patients was provided by one physical therapist trained in standard procedures. The MPT program incorporated two primary components including 1) task-specific training for basic functional tasks and for patient-specific tasks reported to be symptom provoking and 2) progressive strengthening of hip musculature. For the task-specific training, all patients received standard instructions, provided in Appendix A, to optimize their movement pattern during basic daily tasks, such as ascending stairs and sit to stand. Patient-specific tasks were identified by each patient at baseline assessment as symptom-provoking. Using the basic principles of MPT, each patient was instructed in methods to optimize their movement pattern during their identified tasks.

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Most common patient-specific tasks included driving and fitness activities such as running, cycling and swimming. Patients practiced each task using the modifications provided during the visits and were also encouraged to use the task-specific modifications when performing these activities throughout the day.

Strengthening exercises to target specific hip musculature, including hip abductors, external rotators and hip flexors, were also provided. (Appendix B). Patients were instructed to perform these exercises at home on a daily basis. The minimum number of repetitions to be performed at home was determined by the physical therapist based on the performance of patients during the treatment session. They were encouraged to perform the minimum repetitions and increase the number of repetitions as able, as long as they were performing the exercise correctly and experienced no increase in their hip joint pain while performing the exercise.

At each treatment visit, the performance of the functional and patient-specific tasks and strengthening exercises were assessed. Emphasis was placed on independent performance of each task and exercise. For patients to be independent in a given functional task, they must correctly demonstrate the modified lower extremity movement pattern during the task's performance.[15] Once patients were able to demonstrate independent performance of the task as judged by the physical therapist,[15] no further instruction was provided. Patients were encouraged to continue to use the modified movement pattern when performing the task during the day. The tasks reported to be symptom-provoking by each patient at the baseline assessment were prioritized and addressed during the first treatment session. Instruction for the remaining tasks was provided during subsequent visits.

For patients to be independent in a strengthening exercise, they must perform the exercise precisely, demonstrating the appropriate movement and a palpable contraction of the targeted muscle(s). If patients demonstrated independent performance[15] and could complete 2 sets of 20 repetitions, the resistance of the exercise was increased, either by changing body position or adding an elastic resistance band.

Follow-up assessment—Patients in the MPT group returned for follow-up assessment after completion of their treatment. Patients in the Wait-list group returned for follow-up assessment six weeks after baseline testing. The same procedures used during the baseline assessment were used during the follow-up testing.

Outcomes Measures

Primary Outcomes—Retention was defined as the percentage of patients who completed follow-up testing of those enrolled at baseline. Patient adherence to treatment was assessed using attendance to the treatment sessions and weekly patient-reported adherence to their home program. The treating physical therapist's documentation was used to quantify total sessions completed. For adherence to the home exercise program, patients completed a question at each treatment session that asked "On average, how many times per day were you able to complete your exercise program?"

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Secondary Outcomes

Patient-reported outcome measure: The Hip disability and Osteoarthritis Outcome Score (HOOS)[23, 36] is a hip-specific outcome measure developed for people with and without hip osteoarthritis. The HOOS includes 5 subscales to assess 5 domains including Pain, Symptoms, Activities of Daily Living (ADL), Sport and Recreation (Sport) and Quality of Life (QOL). The HOOS has 40 items that are rated on a 5-point Likert scale. Each subscale is transformed to a 100 point scale with higher scores indicating higher levels of function. The HOOS has high test-retest reliability[20, 22, 35] and acceptable content validity.[22, 35] Minimum important change (MIC) for the HOOS subscales has not been established for patients undergoing non-surgical management for CHJP. Using values established for people who have undergone arthroscopic surgery,[22] the MIC values for the HOOS subscales of Pain, Symptoms, ADL, Sport and QOL are 9, 9, 6, 10 and 11, respectively. HOOS data were collected and managed using REDCap (Research Electronic Data Capture) hosted at Washington University.[18]

Kinematics: An 8-camera video-based motion capture system (Vicon, Los Angeles, CA, USA) was used to acquire three-dimensional motion while the patients performed single leg squats. Retro-reflective markers were placed on anatomical landmarks of the pelvis and lower extremities to capture lower limb movement and alignment data. A static calibration trial was collected prior to movement trials. The examiner demonstrated the single leg squat motion to the patient and instructed them to squat as far as they could at their typical pace. For a trial to be valid, the patient had to achieve 60° of knee flexion and maintain their balance. Prior to data collection, each patient was allowed to practice the single leg squat until they were familiar with the movement. Three repetitions of the single leg squat were then collected. If patients demonstrated a loss of balance during a repetition, the repetition was repeated. Visual3D software (C-motion, Inc., Rockville, MD, USA) calculated joint kinematics. The a-priori outcome of interest was hip joint adduction angle at the position of greatest hip flexion, indicating the maximum descent of the single leg squat (Figure 2). A higher value of hip adduction indicated an abnormal movement pattern. Using the described methods to assess seven asymptomatic people, the measure of hip adduction demonstrated excellent within-session reliability ($ICC_{3,3}$ = 0.94), acceptable test-retest reliability ($ICC_{3,3}$ = 0.72) with test-retest standard error of measurement (SEM) of 1.69°.

Muscle Strength: Using previously reported methods,[16] the ERs were tested with the hip flexed to 90° (ERs90°) and 0° (ERs0°) and the ABDs were tested in side lying. Break tests using a *microFET3* hand dynamometer (Hoggan Health Industries, West Jordan, UT) were used to determine maximum muscle force. Three maximal trials were averaged, then multiplied by the associated moment arm of the resistance provided to determine the average torque. Each patient's torque value was normalized by body weight and height to create a body-size independent value.[3] These procedures demonstrated excellent test-retest reliability for the calculated torque values.[16]

Sample size estimation—As the data in this report were obtained to generate preliminary estimates of treatment effectiveness, no a priori power computation was

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performed to determine statistical power to detect between-group differences in treatment effects.

Statistical Analysis

Continuous data are reported as mean ± standard deviation (SD). Ordinal data are reported as median (range). Descriptive statistics were used to characterize the primary outcomes of retention rate and treatment adherence. Retention rates were calculated as the percentage of enrolled patients who completed baseline and follow-up testing and were compared between treatment groups by Fisher's exact test. For treatment session adherence data, we calculated the frequency of patients categorized with at least 80% (≥ five of the six visits) versus less than 80% attendance to treatment sessions. For adherence to the home program, values reported each week were averaged for each patient and the percentage of people that reported completing the home exercise program at least once per day was calculated.

Except where noted, between-group comparison of change between visits for patient-reported outcomes (HOOS), kinematics, and hip muscle strength was performed using analysis of covariance (ANCOVA) where the posttest value was the dependent variable, group was the independent variable, and the pretest value was the covariate. Since residuals were not normally distributed, nonparametric covariance analysis[24] was used for HOOSADL and HOOSSport subscale scores. The heterogeneity of slopes assumption was violated for hip adduction; which indicates that treatment effectiveness for adduction was dependent upon the pretest value of hip adduction. Thus, this outcome was analyzed with a separate slopes ANCOVA where the pretest value was nested within group. This outcome was tested at each pretest decile and quartile, with specific data reported at the two levels which discriminate between group differences that are statistically significant (i.e., 21.5° for the 60th percentile) and are not significant (i.e., 19.6° for the 50th percentile). As homogeneity of slopes was confirmed for all other outcomes, treatment effectiveness for all other outcomes was tested independent of the pretest value. For each outcome, the adjusted treatment effect estimate is reported to reflect the between-group difference at posttest from the ANCOVA model least square means after adjusting for the pretest value.

Data were analyzed using SAS software, version 9.3 of the SAS system for Linux (SAS Institute Inc., Cary, NC, USA). P value < 0.05 was considered statistically significant. The Consolidated Standards of Reporting Trials Statement (CONSORT) was used for reporting.[42]

RESULTS

Of the 102 potential patients screened for eligibility, 35 were enrolled and participated in baseline testing and randomization (Figure 1). Main reasons for exclusions were age greater than 40 years (27%), pain not located in hip joint or groin (22%), BMI greater than 30 (18%), and knee or low back pain limiting mobility (10%). Other reasons (22%) included pain less than 3/10, previously received MPT, diagnosed hip osteoarthritis or hip surgery, and contraindication to MRI. Group characteristics for those enrolled and those who provided posttest data are provided in Table 1. Eighteen patients were randomized into the MPT group and 17 patients were randomized to the Wait-list group. No adverse events were

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reported during the study. The median number of days between baseline and follow-up assessment were greater in the MPT group (64.5 [range 52–96]) compared to the Wait-list group (52.2 [range 40–90], $P = 0.001$)

Primary Outcomes

Retention rates were similar for the MPT and the Wait-list groups (89% (16/18) and 94% (16/17), p = 1.0), respectively. The HOOS data for one patient and kinematic data for another were lost due to technical issues. Sixteen of the 18 patients (89%) in the MPT group attended at least 80% of the treatment sessions; 15 patients attended all six sessions and one attended five sessions. For the home exercise program, 89% of patients reported performing their home program at least once per day.

Secondary Outcomes (Table 2)

Patient-Reported Outcome—Compared to the Wait-list group, the MPT group reported higher scores, indicating better function in the HOOS subscales for Symptoms, ADL, and Sport at the follow-up visit. There were no statistically significant between-group differences in HOOS subscales for Pain or QOL.

Kinematics—On average, the MPT group demonstrated a reduction in hip adduction during the single leg squat at the follow-up visit (−2.6° ± 7.6°) and the Wait-list group demonstrated a slight increase (0.8° ± 4.6°). However, treatment effectiveness for this outcome at posttest depended upon the pretest level of hip adduction. As such, treatment effectiveness for this outcome was tested at pretest values corresponding to each decile and quartile. There was a statistically significant between-group difference in the amount of hip adduction at posttest, but only for patients with higher hip adduction at pretest. When comparing groups, the MPT group improved significantly more at all pretest adduction levels we tested that were 21.5° or greater; meaning that when comparing groups in patients whose pretest hip adduction is at or above the 60th percentile (or ≥ 21.5°), MPT showed significant improvement compared to those in the Wait-list. No significant treatment differences were found for patients with less hip adduction at pretest.

Hip Muscle Strength—There were no statistically significant between-group differences in hip muscle strength of hip abductors ($P = 0.21$) and hip external rotators ($P \geq 0.13$).

DISCUSSION

To our knowledge there are no clinical trials in the literature regarding the use of MPT to reduce symptoms and improve function in people with CHJP. Feasibility studies play an important role in the planning of RCTs for novel interventions or application of existing interventions in new patient populations or recruitment settings.[1] We demonstrated excellent retention rates for those patients who received MPT as well as those who were randomized to the Wait-list group. Patient adherence to supervised sessions and home program performance was also high. Secondary outcomes suggest that the MPT may result in improvements in patient-reported function and lower extremity movement patterns, even in

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patients reporting a median of two years of pain. Completion of this study has confirmed the feasibility of mounting a larger trial to examine treatment effects more precisely.

Retention rate for the MPT and Wait-list group was 89% and 94% respectively, which exceeded our hypothesis and recommendations suggesting a retention rate greater than 80% to avoid serious threats to validity.[43] We provided compensation for each testing session and treatment was provided free of charge, which may contribute to high retention rates in both groups. Among the 16 patients who completed treatment, adherence to the supervised sessions and home exercise performance was high. One factor that may have contributed to our patients' high treatment adherence was the education provided to patients during their treatment sessions.[30] A key component to the task-specific training was educating patients in basic biomechanical principles of movement and the theoretical concept that modifying their movement pattern during daily and patient-specific activities may result in reduced stresses to joint structures and thus reduce their pain. Our standardized procedure[15] to assess the performance of the functional tasks and strengthening exercises may also facilitate treatment adherence. Patients are more likely to adhere to exercises they feel confident in performing and that do not increase their pain.[5] Both task-specific training and strengthening exercises, were evaluated at each visit for proper performance and to determine the appropriate level of difficulty. If a task or an exercise was too challenging or pain-provoking, the physical therapist would provide modifications prior to prescribing the home program.

Our study was designed to assess feasibility; therefore we did not expect to find statistical differences in our secondary outcome measures. Nevertheless, some treatment effect sizes were greater than anticipated. Statistically significant differences exceeding MIC were found in HOOS Symptoms, ADL, and Sport subscales, with the MPT group demonstrating greater improvement at follow up compared to the Wait-list group. The MPT group also demonstrated a reduction in hip adduction motion that was greater than measurement error, however the clinical significance of this reduction is unknown. The between-group difference in hip adduction motion was primarily attributed to those patients who demonstrated baseline hip adduction angles greater than 21°, suggesting a subgroup of patients may benefit more from MPT. Our study was not powered on the hip adduction variable; therefore we cannot make a definitive conclusion. Investigation of this relationship is a goal of our future work.

Surprisingly, no significant differences were noted in posttest hip muscle strength, raising questions about dosage and the relationship among hip strengthening, hip-specific function and lower extremity kinematics in people with CHJP. The six week duration may not allow sufficient time for people with CHJP to achieve large improvements in muscle strength. Hip muscle performance may be compromised in our patients due to their hip joint injury and potential arthrogenic neuromuscular inhibition.[14] Approximately 50% of those participating in treatment were unable to reach higher levels of strengthening exercises proposed in the study, suggesting that longer treatment duration may be needed to improve muscle strength when the hip joint is symptomatic. Importantly, patients receiving MPT demonstrated an increase in patient-reported function and improvement in lower extremity movement pattern, despite no improvement in hip muscle strength. The improvement in function may be due to

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the task-specific training to modify functional activities, particularly those activities reported to be symptom-provoking.

Based on our results, an RCT to compare the effectiveness of MPT to a standard rehabilitation approach would be feasible. Our methods for retention and patient adherence are appropriate and will be used in the future trial. We will modify the MPT protocol to include task-specific instruction only. Treatment will include repeated practice of functional tasks using optimized movement patterns. The functional tasks will be progressed in difficulty based on the patient's performance. Using the best evidence available,[11, 12, 45] treatment in the standard arm will focus on reversing impairments of lower extremity flexibility and lower extremity and trunk strength. To optimize potential strength changes in the standard arm and movement pattern changes in the MPT arm, we will lengthen the treatment duration to 12 weeks. Our proposed study design will allow us to compare the independent effect of the task-specific instruction and hip muscle strengthening on important patient outcomes.

The outcome variables used in this feasibility study will be used in the future trial. Our primary outcome of interest, the HOOS, will be used to measure patient reported hip-specific function. Using the preliminary data generated by this feasibility study for HOOS Symptoms, a sample size computation was performed for a future trial that compares the change in response to MPT versus standard treatment. Since the future trial will implement longer treatment duration, estimates of retention and variability are expected to differ from those observed in the feasibility study. Conservatively assuming that (a) 85% of participants will complete post-treatment testing; (b) the common within-group standard deviation for change between pre- and immediate post-treatment is 15; and (c) no covariates will be informative in the analysis, a sample size of 52 participants per group (with 45 participants per group completing both visits) will have 80% statistical power to detect a 9-point MIC with a two-tailed t-test at alpha 0.05. For kinematic assessment, we will focus on the single leg squat, however will assess other tasks such as step down,[9, 26] sit to stand[10] and when appropriate a drop vertical jump.[9] Finally, we will assess outcomes one year after treatment to assess the sustainability of treatment effects.

Limitations

Our study was designed to assess the feasibility of implementing MPT for people with CHJP; therefore, we cannot make definitive statements regarding the efficacy of MPT. We enrolled patients with CHJP, which may be considered a heterogeneous sample; however our sample is representative of people who are seen in physical therapy clinics. Because we used MRI in the parent study, patients who had contraindications to MRI or were pregnant were not enrolled. Only one patient was unable to participate in the study due to these exclusion criteria. We included a control group in this study in order to determine if MPT would result in better outcomes than no treatment. The Wait-list group received no additional attention between the baseline and follow-up testing session; therefore, we cannot be certain that MPT is superior to attention alone. Additionally, we cannot state that MPT is superior to other rehabilitation strategies. We do not know if strength improvements occurred in the iliopsoas. In our clinical experience, maximum strength testing of iliopsoas is often symptom-

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provoking, therefore we chose not to assess iliopsoas strength. Time to return for follow-up was longer for the MPT group. Some patients required longer than six weeks to complete all six visits, due to scheduling issues or personal reasons. Finally, an important component of MPT is the assumption that patients will incorporate the movement pattern modifications into their daily life. The patients were encouraged to practice their functional tasks and to also use the movement pattern modifications as they performed their daily tasks. Currently, we have no objective method to assess the patients' ability to implement the movement pattern modifications in their daily routine. Future technological advances in body worn sensors may improve our capability to measure movement in non-laboratory situations.

CONCLUSION

Based on our retention and adherence rates, a larger RCT appears to be warranted and feasible. Although this study was designed to assess feasibility only, we found statistically significant between-group differences in hip-specific function and lower extremity kinematics. Our findings highlight potential modifications for the future trial, such as modification to the MPT protocol and increasing treatment duration. Completion of a future, larger trial comparing MPT to standard treatment will determine the efficacy of MPT and improve our understanding of the factors associated with treatment outcomes, ultimately leading to improved strategies for people with CHJP.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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KEY POINTS

Findings

A clinical trial to compare the effectiveness of MPT to a standard rehabilitation approach for patients with CHJP is feasible as demonstrated by high retention and patient adherence rates.

Implications

Rehabilitation using MPT may yield positive outcomes for people with CHJP. Future study to assess the effectiveness of MPT is needed.

Caution

The study was designed to assess feasibility; therefore the sample size is small. Movement pattern training resulted in improved outcomes compared to no treatment; however we cannot conclude that MPT is superior to attention alone or to other treatment strategies.

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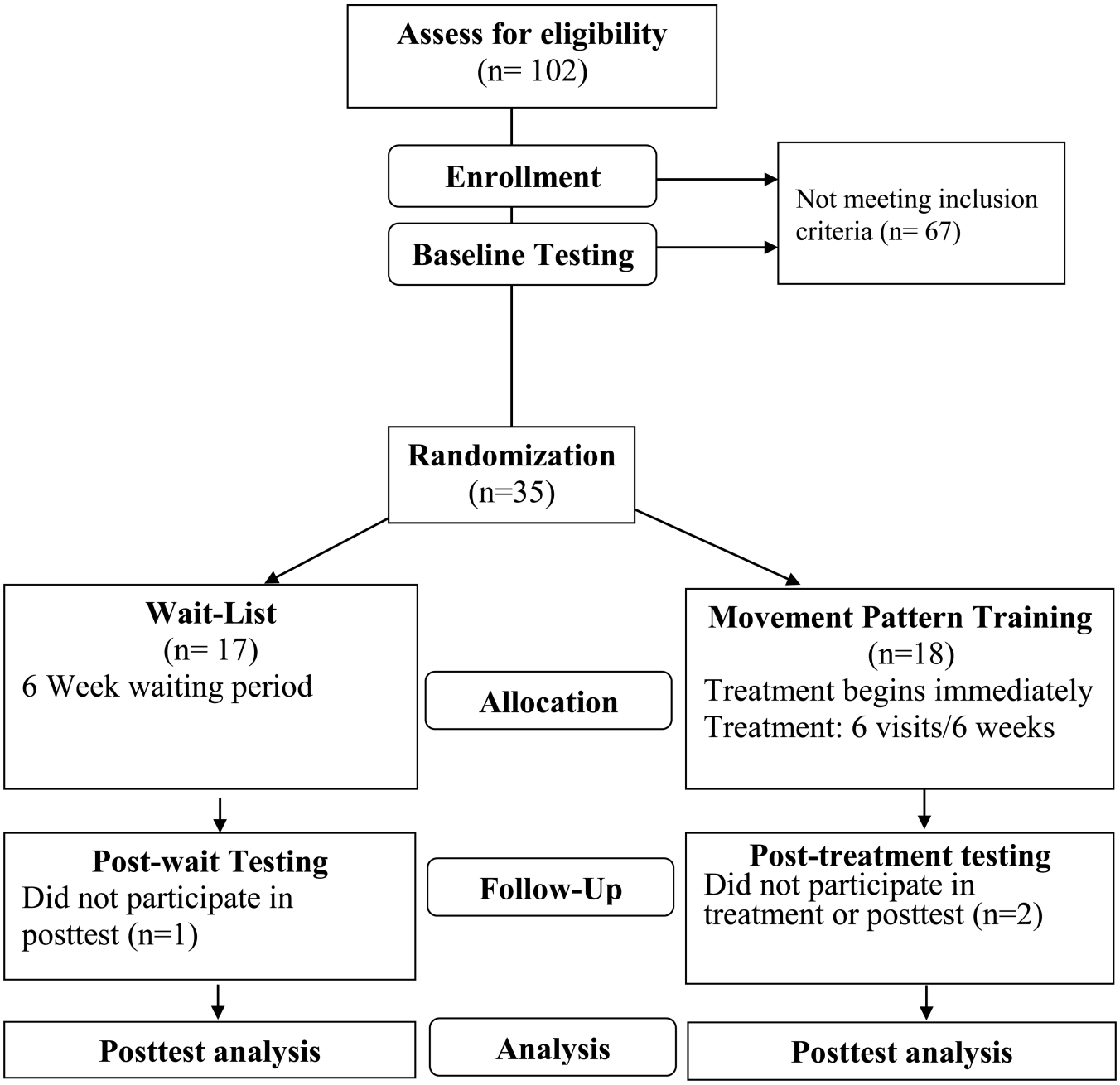

Figure 1.

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Figure 2.

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TABLE 1

Demographics for all enrolled participants and by treatment group for patients who provided posttest data.

Variable	All Enrolled Patients N=35	Patients Providing Posttest Data		
		All N=32	MPT N=16	Wait-list N=16
Demographics				
Sex	29F:6M	27F:5M	15F:1M	12F:4M
Limb side	19R:16L	17R:15L	9R:7L	8R:8L
Pain involved side	13R:9L:13B	11R:8L:13B	6R:3L:7B	5R:5L:6B
Age, years *	28.3±5.2	28.1±5.0	27.2±5.1	29.0±4.8
BMI, kg/m^2 *	24.4±2.9	24.3±2.8	24.2±3.0	24.4±2.6
UCLA †	9 (3–10)	9 (3–10)	9 (4–10)	7.5 (3–10)
Pain report				
Duration, years ‡	2.0 (0.4–10)	2.0 (0.4–10)	2.0 (0.5–10)	1.5 (0.4–10)
Avg pain ‡	3.5 (1–8)	3.0 (1–8)	3.5 (1–8)	3.0 (1–7)
Worst pain ‡	6.0 (2–10)	6.0 (2–10)	6.0 (2–10)	6.0 (3–10)

Abbreviations: MPT, movement pattern training group; F, female; M, male; R, right; L, left; B, bilateral; BMI, body mass index; UCLA, University of California Los Angeles Activity Score; Avg, average

*Values are mean ± SD.

†Values are median (range). UCLA: patients are asked to rate their activity level over the previous 6 months. 0=wholly inactive, dependent on others; 10= regularly participates in impact sports.

‡Values are median (range). Pain rated by patients using a verbal numerical pain rating scale. 0=no pain; 10=worst pain imaginable

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TABLE 2

Summary of results for secondary outcome variables.

Variable	Pretest Mean ± SD	Posttest Mean ± SD	Within-Group change* Mean ± SD	Adjusted treatment effect† Mean (95% CI)	*P* Value§
Patient-reported outcomes					
HOOSPain‡					
MPT	78.2 ± 12.3	81.5 ± 13.2	3.3 ± 9.9	2.7 (−4.5, 9.9)	0.45
Wait-list	73.9 ± 13.8	75.8 ± 13.4	1.9 ± 10.4		
HOOSSymptoms‡					
MPT	75.0 ± 17.0	85.0 ± 13.6	10.0 ± 9.3	12.8 (3.1, 22.5)	0.01
Wait-list	68.4 ± 15.8	67.2 ± 20.9	−1.3 ± 16.1		
HOOSADL‡					
MPT	90.7 ± 9.9	93.5 ± 10.9	2.8 ± 7.6	8.6 (1.2, 16.1)	0.02
Wait-list	90.6 ± 10.2	84.8 ± 14.1	−5.8 ± 11.8		
HOOSSport‡					
MPT	77.1 ± 17.5	84.6 ± 19.6	7.5 ± 15.4	9.4 (0.1, 18.8)	0.048
Wait-list	70.3 ± 23.3	70.7 ± 16.3	0.4 ± 14.3		
HOOSQOL‡					
MPT	65.1 ± 13.0	71.3 ± 18.3	6.2 ± 10.6	−1.0 (−9.7, 7.8)	0.82
Wait-list	55.3 ± 19.2	63.1 ± 19.7	7.8 ± 11.8		
Hip Kinematics (°)					
Flexion					
MPT	67.4 ± 14.0	61.7 ± 15.9	−5.7 ± 14.7	−8.8 (−19.4, 1.8)	0.099
Wait-list	75.4 ± 11.7	73.7 ± 12.9	−1.8 ± 16.2		
Adduction					
MPT	20.2 ± 6.6	17.6 ± 5.7	−2.6 ± 7.6	−4.1 (−8.2, −0.1) at 21.5° pretest//	0.049 at 21.5° pretest//
Wait-list	18.9 ± 6.2	19.7 ± 7.1	0.8 ± 4.6	−2.8 (−6.7, 1.1) at 19.6° pretest//	0.15 at 19.6° pretest//
Internal Rotation					
MPT	2.5 ± 7.7	2.7 ± 6.3	0.2 ± 7.3	1.9 (−2.1, 5.9)	0.33

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Variable	Pretest Mean ± SD	Posttest Mean ± SD	Within-Group change* Mean ± SD	Adjusted treatment effect† Mean (95% CI)	*P* Value§
Wait-list	5.7 ± 6.8	2.2 ± 5.9	−3.5 ± 5.9		
Hip Muscle Strength¶					
ABDs					
MPT	6.7 ± 1.8	7.2 ± 2.3	0.6 ± 1.8	0.7 (−0.4, 1.8)	0.21
Wait-list	7.2 ± 2.3	7.0 ± 2.1	−0.2 ± 1.2		
ERs90°					
MPT	3.4 ± 0.8	3.5 ± 0.7	0.1 ± 0.8	−0.3 (−0.7, 0.2)	0.23
Wait-list	3.6 ± 0.8	3.9 ± 0.9	0.3 ± 0.4		
ERs0°					
MPT	2.9 ± 0.9	3.0 ± 1.0	0.1 ± 0.7	−0.3 (−0.7, 0.1)	0.13
Wait-list	2.8 ± 0.7	3.2 ± 0.8	0.4 ± 0.4		

Abbreviations: SD, standard deviation; CI, confidence interval; MPT, movement pattern training group; HOOS, Hip Disability and Osteoarthritis Outcome Score; HOOSADL, function in activities of daily living; HOOSSport, function in sports and recreation; HOOSQOL, quality of life; ABDs, abductors with the hip abducted 15°; ERs90°, external rotator strength with hip in 90° flexion; ERs0°, external rotator strength with hip in neutral flexion/extension.

*Change is calculated by subtracting the pretest value from the posttest value.

†Adjusted treatment effects are calculated by subtracting the MPT minus Wait-list.

‡Patient-reported outcome measures with 100=no disability.

§P value by analysis of covariance (ANCOVA).

//Adjusted treatment effects and p-values are reported at the two pretest levels which discriminate between group differences that are and are not statistically significant.

¶Muscle torque (Nm) was normalized by body weight (N) × height (m) × 100.